Two-Dimensional Sampling Theorem

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Key Concepts

1) Sampling in two dimensions (2D) converts a continuous-space image to a discrete-space image consisting of pixels.

2) If sampling intervals (lengths) are measured in units of meters

    a) Continuous-space radian frequency is measured in units of rads/meter.

    b) Discrete-space frequency is measured in units of rads.

3) Sampling in 2D space introduces replication (aliasing) in the 2D frequency domain.

4) Suppose B_x and B_y (rads/meter) are the highest horizontal and vertical spatial frequencies in the continuous-space image. Then, we require the sampling intervals to satisify $\Delta x \leq \frac{2\pi}{2B_x}$ and $\Delta y \leq \frac{2\pi}{2B_y}$ (meters) to avoid aliasing and uniquely represent the continuous-space image with samples.

2-D Sampling

$f[m,n] = f(m\Delta x, n\Delta y)$ sampling intervals $\Delta x, \Delta y$ meters

Continuous-space frequency
U, V rads/meter

Discrete-space frequency
u, v rads

$$u = U\Delta x$$
$$v = V\Delta y$$

Sampling frequencies

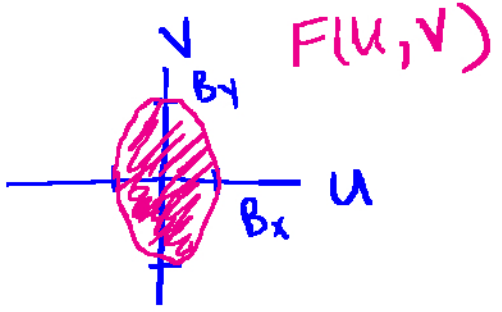
$$U_s = \frac{2\pi}{\Delta x} \text{ rads/meter}$$
$$V_s = \frac{2\pi}{\Delta y} \text{ rads/meter}$$

$$f_s(x,y) = \sum_{m,n} f[m,n]\,\delta(x-m\Delta x, y-m\Delta y)$$
$$= f(x,y) \underbrace{\sum_{m,n} \delta(x-m\Delta x, y-m\Delta y)}_{s(x,y)}$$

$\xleftrightarrow{FT}$

$$F_s(U,V) = F(U,V) * S(U,V)$$
$$\propto F(U,V) * \sum_{k,l} \delta(U-kU_s, V-lV_s)$$
$$= \sum_{k,l} F(U-kU_s, V-lV_s)$$

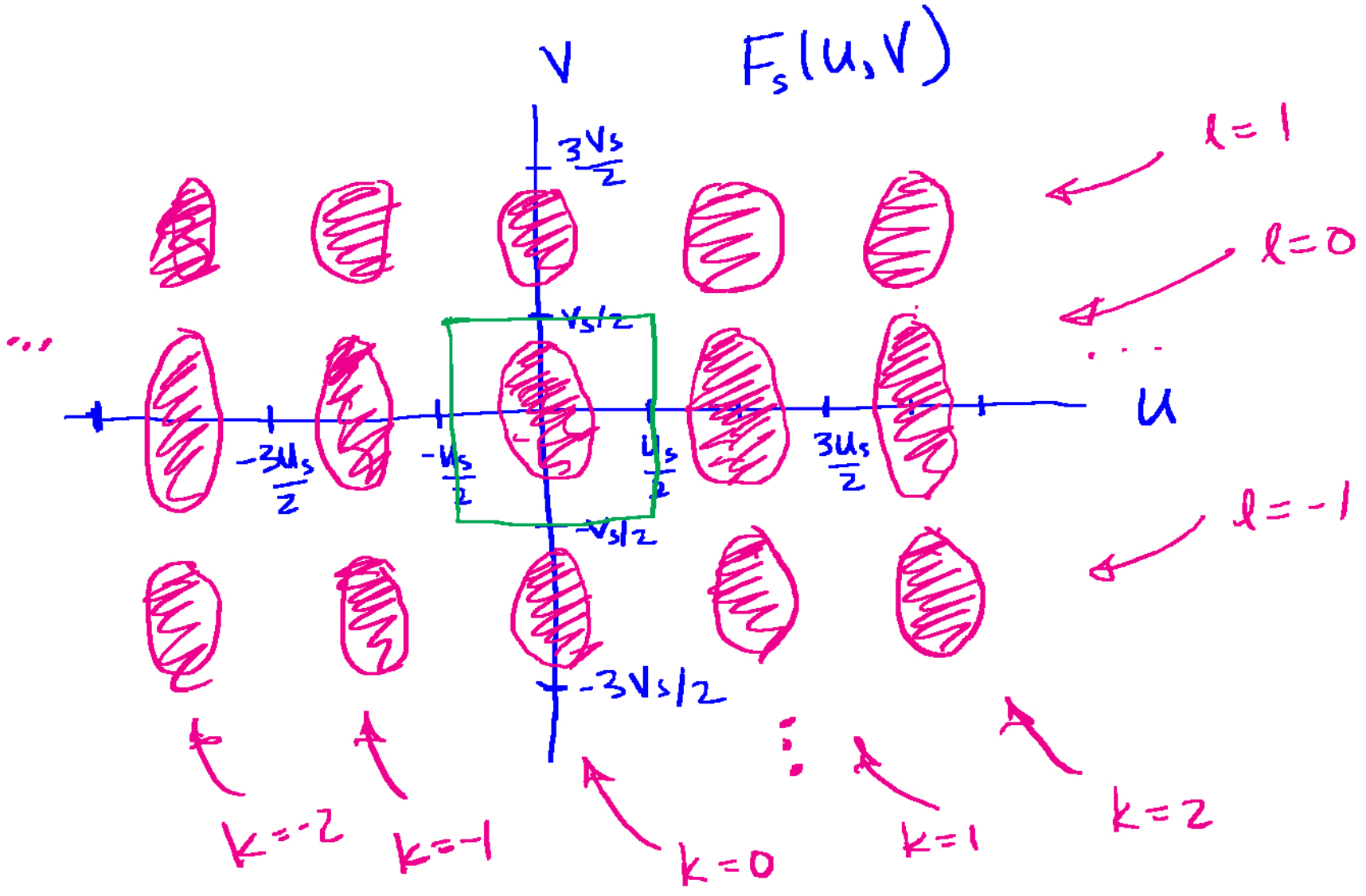
$$f_s(x,y) = \sum_{m,n} f[m,n]\,\delta(x - m\Delta x,\ y - n\Delta y) \quad \overset{FT}{\longleftrightarrow} \quad F_s(u,v) \propto \sum_{k,\ell} F(u - k u_s,\ v - \ell v_s)$$



For no aliasing

$$u_s = \frac{2\pi}{\Delta x} > 2B_x$$

$$v_s = \frac{2\pi}{\Delta y} > 2B_y$$



Nyquist Sampling Theorem (2D)

Let $f(x,y)$ be bandlimited to $-B_x < u < B_x, -B_y < v < B_y$

If $f[m,n] = f(m\Delta x, n\Delta y)$ and $\Delta x < \frac{\pi}{B_x}, \Delta y < \frac{\pi}{B_y}$,

then $f(x,y)$ can be uniquely recovered from the samples $f[m,n]$.

